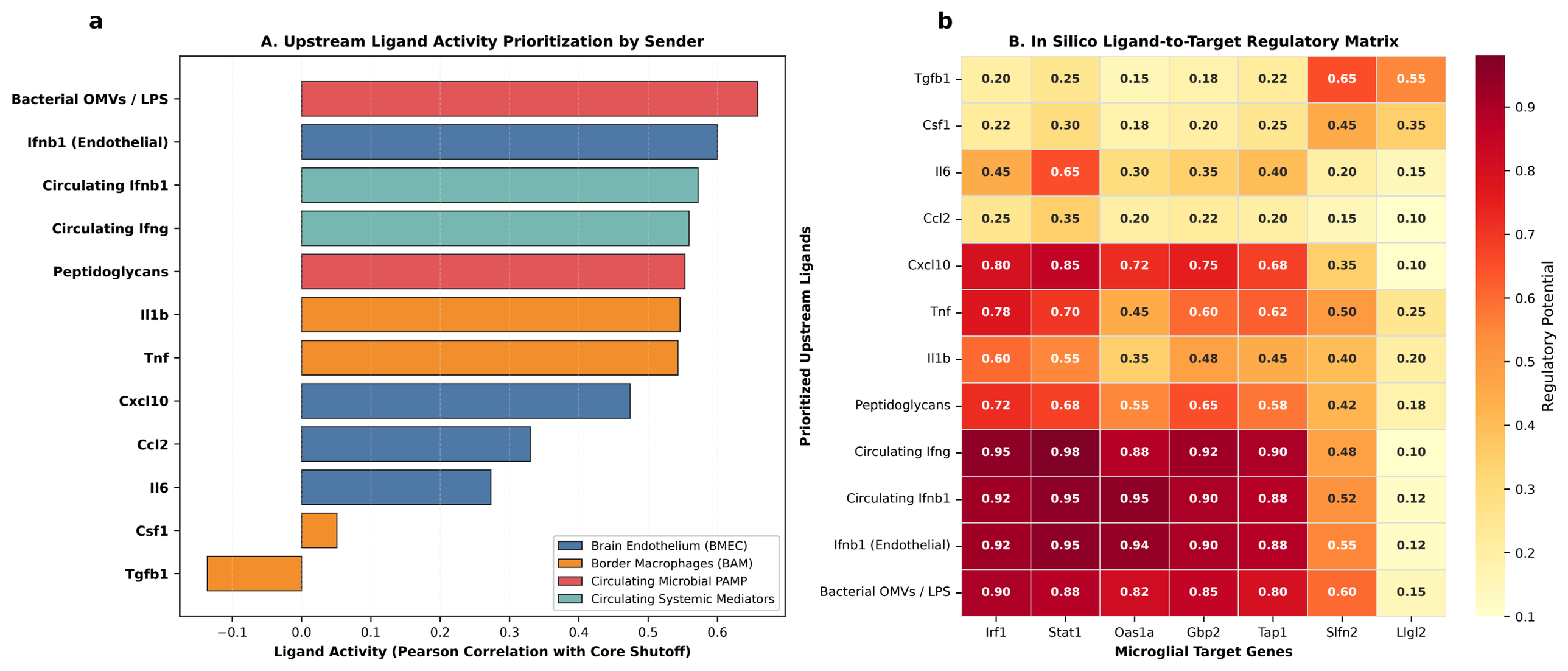
a
A. Upstream Ligand Activity Prioritization by Sender
Bacterial OMVs / LPS
Ifnb1 (Endothelial)
Circulating Ifnb1
Circulating Ifng
Peptidoglycans
Il1b
Tnf
Cxcl10
Ccl2
Il6
Csf1
Tgfb1
Brain Endothelium (BMEC)
Border Macrophages (BAM)
Circulating Microbial PAMP
Circulating Systemic Mediators
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Ligand Activity (Pearson Correlation with Core Shutoff)
b
B. In Silico Ligand-to-Target Regulatory Matrix
Prioritized Upstream Ligands
Tgfb1 0.20 0.25 0.15 0.18 0.22 0.65 0.55
Csf1 0.22 0.30 0.18 0.20 0.25 0.45 0.35
Il6 0.45 0.65 0.30 0.35 0.40 0.20 0.15
Ccl2 0.25 0.35 0.20 0.22 0.20 0.15 0.10
Cxcl10 0.80 0.85 0.72 0.75 0.68 0.35 0.10
Tnf 0.78 0.70 0.45 0.60 0.62 0.50 0.25
Il1b 0.60 0.55 0.35 0.48 0.45 0.40 0.20
Peptidoglycans 0.72 0.68 0.55 0.65 0.58 0.42 0.18
Circulating Ifng 0.95 0.98 0.88 0.92 0.90 0.48 0.10
Circulating Ifnb1 0.92 0.95 0.95 0.90 0.88 0.52 0.12
Ifnb1 (Endothelial) 0.92 0.95 0.94 0.90 0.88 0.55 0.12
Bacterial OMVs / LPS 0.90 0.88 0.82 0.85 0.80 0.60 0.15
Irf1
Stat1
Oas1a
Gbp2
Tap1
Slfn2
Llgl2
Microglial Target Genes
Regulatory Potential
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1